

Module 12: Final Project Workbench

Complete system, evaluation report, and demo

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Primary sources: Course synthesis from Osmani; Thomas & Hunt project practices.

Learning Outcomes

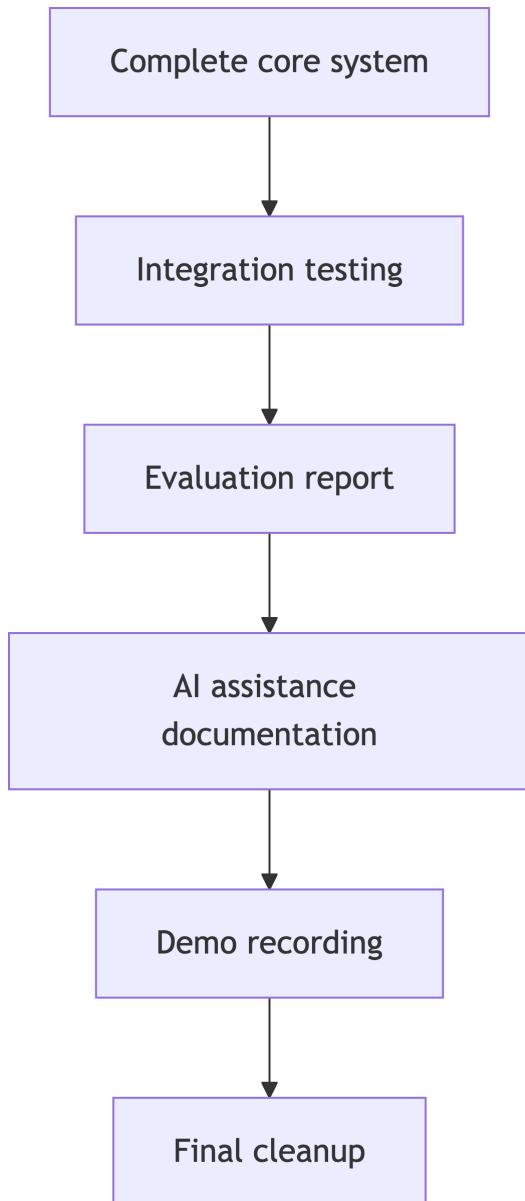
- Design and implement a complete AI-assisted system.
 - Integrate workflows, evaluation, and safeguards.
 - Justify architectural and engineering decisions.
 - Present and defend the system.
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Final System Expectations

Your Controlled Coding Agent should demonstrate:

- Structured input handling.
 - AI-assisted generation.
 - Evaluation and testing.
 - Controlled workflow management.
 - Persistence or artifact tracking.
 - Documentation.
 - Human oversight.
 - Collaborative engineering practice.
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Final Submission Timeline



Reproducibility

A grader should be able to answer:

- How do I set it up?
 - How do I run it?
 - What example should I try?
 - What output should I expect?
 - How do I run tests?
 - Where are evaluation results?
 - Where is AI assistance documented?
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Evaluation Report

The evaluation report should include:

- What was evaluated.
 - Metrics or criteria.
 - Test cases or scenarios.
 - Results.
 - Interpretation.
 - Known limitations.
 - What would improve next.
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AI Assistance Section

Your README should make AI use transparent:

- Planning.
- Code generation.
- Test generation.
- Debugging.
- Review.
- Integration support.
- Documentation support.
- Accepted, modified, and rejected suggestions.
- Verification methods.

Demo Structure

Recommended 5-8 minute flow:

1. Problem and users.
 2. Architecture overview.
 3. Live or recorded workflow run.
 4. Evaluation results.
 5. Safeguards and human oversight.
 6. Team process and AI evidence.
 7. Lessons learned.
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Professional Defense

Be ready to explain:

- Why this architecture?
 - What did AI help with?
 - What did humans correct?
 - What risks remain?
 - Which tests matter most?
 - What would you do next with more time?
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Final Quality Checklist

- Clean repository organization.
 - No committed secrets.
 - Fresh-environment setup tested.
 - Tests and examples run.
 - Demo link included.
 - Required document paths submitted.
 - Commit hash captured.
 - Individual contributions clear.
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Course Synthesis

Responsible AI-assisted engineering means:

- Clear intent.
 - Structured prompts.
 - Human review.
 - Automated evidence.
 - Risk awareness.
 - Transparent documentation.
 - Team accountability.
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Closing Reflection

Answer individually:

- What AI interaction changed your process most?
 - What risk did you underestimate early?
 - What verification habit will you keep?
 - What work should remain human-led?
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Takeaways

- The final project is a demonstration of engineering judgment.
- AI assistance is strongest when controlled, evaluated, and documented.
- The goal is not to show that AI wrote code; it is to show that your team built trustworthy software with AI assistance.